

IEEE 14-Bus GFL↔GFM Switching

Corrected full-state equations, automatic formation, and the requested 200-s chronology

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Abstract

This report describes the corrected project-owned model used for one six-state EMF6 synchronous generator (SG) and four switchable inverter-based resources (IBRs). The correction restores the physical controller ownership: grid-following (GFL) mode contains a phase-locked loop (PLL), while grid-forming (GFM) mode forms its angle through a virtual swing equation and contains no PLL. Both modes share the AC-filter current and dc-link states. The requested chronology uses $\Delta t = 0.1$ s and events at 20, 50, 85, 85.15, 110, and 145 s, with a requested endpoint of 200 s. The latest generated accepted trajectory ends at 200.000 s. Its observed network-voltage range is 0.014062–2.406517 pu and its aggregate-frequency range is 55.390979–84.329308 Hz. These extrema are reported as results, not used as tuned acceptance bands. The maximum accepted residual is 9.9395×10^{-9} , the deepest step subdivision is 6, and the number of rejected domain trials is 0. The SG re-closed at 154.300 s and the four independent handback transactions reached all-GFL at 194.100 s. Numerical completion is established; the large reported excursions explicitly preclude an operational-readiness claim.

1 Scope, evidence classes, and non-goals

The system base is $S_{base} = 100$ MVA and $f_0 = 60$ Hz. SG1 is connected at bus 1; IBR1–IBR4 are connected at buses 2, 3, 6, and 8. Network equations, power flow, equilibrium, small-signal analysis, and time-domain simulation use project-owned Base MATLAB code. The model choices are separated as follows.

Class	Use in this work
Source-defined / source-mapped	AC-filter, GFL PLL and PI loops, GFM virtual swing and voltage-amplitude dynamics, controller gains, and event chronology.
Case-defined	Bus/branch data, resource ratings and limits, initial P/Q schedule, E_{ref} , event times, and healthy per-bus voltage references.
Project-derived	Fixed dual-mode state superset, dc-source closure, mode-transfer map, lost-SG dispatch participation, equilibrium gauge, selector ranking, grid-relative SG synchronizer, and subdivision-history initialisation.
Numerical method	Newton solves, finite-difference Jacobians, one rotational quotient coordinate, implicit trapezoidal integration, and event-local backward-Euler half steps.

The run does not claim EMT accuracy, unbalanced-fault validity, or controller field readiness. In particular, a transient voltage excursion is disclosed rather than clipped, filtered, or used to retune the equations.

2 Corrected fixed state and input contracts

Every IBR stores one 20-state vector of constant dimension:

$$x = [i_d, i_q, V_{dc}, \delta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q, \xi_{Id}^{gfl}, \xi_{Iq}^{gfl}, V_{d,del}^{gfl}, V_{q,del}^{gfl}, \theta, \omega, E, \xi_{Vd}, \xi_{Vq}, \xi_{Id}^{gfm}, \xi_{Iq}^{gfm}, V_{d,del}^{gfm}, V_{q,del}^{gfm}]^T. \quad (1)$$

The common plant is $x_{1:3}$. GFL activates $x_{1:11}$ and freezes $x_{12:20}$ exactly. GFM activates $x_{1:3}$ and $x_{12:20}$ and freezes $x_{4:11}$ exactly. Hence GFL has 11 active states including its PLL, while GFM has 12 active states including its VSG angle, frequency, and voltage-amplitude states but no PLL. The fixed external input vector is

$$u = [P_{ref}, Q_{ref}, E_{ref}]^T. \quad (2)$$

GFL consumes P_{ref}, Q_{ref} . GFM consumes all three inputs; E_{ref} is an external terminal-voltage reference, not the measured $|V|$.

2.1 Per-unit boundary and terminal power

Let $\kappa = S_{base}/M_{base}$ and let ϑ be the active controller angle (δ_{PLL} in GFL or θ in GFM). The model uses the power-invariant project boundary

$$I_{sys} = \frac{(i_d + j i_q) e^{j\vartheta}}{\kappa}, \quad S_{sys} = P + jQ = V I_{sys}^*. \quad (3)$$

No isolated factor $3/2$ is inserted in only part of the per-unit model. This identity is also the independent current/power oracle used by tests and reporting adapters.

3 Common converter plant and dc closure

With $V e^{-j\vartheta} = v_d + j v_q$, limited current $i_d + j i_q$, filter parameters L_f, R_f , and per-unit frequency ω_c , the common current plant is

$$\frac{L_f}{\omega_b} \dot{i}_d = V_{d,del} - v_d - R_f i_d + \omega_c L_f i_q, \quad (4)$$

$$\frac{L_f}{\omega_b} \dot{i}_q = V_{q,del} - v_q - R_f i_q - \omega_c L_f i_d. \quad (5)$$

The circular limit acts on the controller current reference only. If $i^{*,raw} = i_d^{*,raw} + j i_q^{*,raw}$, then

$$i^* = \alpha i^{*,raw}, \quad \alpha = \min\left(1, \frac{I_{max}}{|i^{*,raw}|}\right). \quad (6)$$

The physical filter states i_d, i_q and the terminal-current injection are never projected, clipped, or replaced by i^* . Consequently a fast fault transient may drive actual filter current above the

reference limit; the report discloses that response rather than altering the plant. The command delay is

$$\dot{V}_{d,del} = \frac{V_{d,cmd} - V_{d,del}}{T_d}, \quad \dot{V}_{q,del} = \frac{V_{q,cmd} - V_{q,del}}{T_d}. \quad (7)$$

The physical energy balance requires an external dc current, but the case does not prescribe its controller. The project therefore uses the declared exogenous controlled-current closure

$$I_{dc} = \frac{P_{ac}}{V_{dc}} + \frac{C_{dc}}{T_{dc}}(V_{dc,ref} - V_{dc}), \quad (8)$$

$$C_{dc}\dot{V}_{dc} = I_{dc} - \frac{P_{ac}}{V_{dc}}, \quad P_{ac} = V_{d,cmd}i_d + V_{q,cmd}i_q, \quad (9)$$

$$\therefore \dot{V}_{dc} = \frac{V_{dc,ref} - V_{dc}}{T_{dc}}. \quad (10)$$

Instantaneous power feed-forward prevents this unpublished closure from moving the sourced AC equilibrium. Nonpositive or nonfinite V_{dc} fails closed.

4 GFL equations: PLL present

The GFL frame is owned by its PLL:

$$v_d + jv_q = Ve^{-j\delta_{PLL}}, \quad (11)$$

$$\dot{\xi}_{PLL} = v_q, \quad (12)$$

$$\dot{\delta}_{PLL} = k_{p,PLL}v_q + k_{i,PLL}\xi_{PLL}, \quad (13)$$

$$\omega_{PLL} = 1 + \dot{\delta}_{PLL}/\omega_b. \quad (14)$$

Measured inverter-base power is $P_{inv} = \kappa P$ and $Q_{inv} = \kappa Q$. The outer loops are

$$e_P = \kappa P_{ref} - P_{inv}, \quad \dot{\xi}_P = e_P, \quad i_d^* = k_{pP}e_P + k_{iP}\xi_P, \quad (15)$$

$$e_Q = \kappa Q_{ref} - Q_{inv}, \quad \dot{\xi}_Q = e_Q, \quad i_q^* = -(k_{pQ}e_Q + k_{iQ}\xi_Q). \quad (16)$$

After circular limitation of $i_d^* + ji_q^*$, the inner current errors are $e_d = i_d^* - i_d$ and $e_q = i_q^* - i_q$, and

$$V_{d,cmd} = v_d + R_f i_d - \omega_{PLL} L_f i_q + k_{pI} e_d + k_{iI} \xi_{Id}, \quad (17)$$

$$V_{q,cmd} = v_q + R_f i_q + \omega_{PLL} L_f i_d + k_{pI} e_q + k_{iI} \xi_{Iq}. \quad (18)$$

Let $r_d = i_d^{*,raw} - i_d^*$ and $r_q = i_q^{*,raw} - i_q^*$. Anti-windup belongs to the outer P/Q integrators because their output is the limited current reference:

$$\dot{\xi}_P = \begin{cases} 0, & \alpha < 1 \text{ and } k_{iP}e_P r_d > 0, \\ e_P, & \text{otherwise,} \end{cases} \quad (19)$$

$$\dot{\xi}_Q = \begin{cases} 0, & \alpha < 1 \text{ and } (-k_{iQ})e_Q r_q > 0, \\ e_Q, & \text{otherwise.} \end{cases} \quad (20)$$

The inner current controller has no voltage-command limiter in this model, hence $\dot{\xi}_{Id} = e_d$ and $\dot{\xi}_{Iq} = e_q$ without the outer-loop hold condition.

5 GFM equations: VSG angle, no PLL

GFM owns an absolute virtual-machine angle and frequency:

$$\dot{\theta} = \omega_b(\omega - 1), \quad (21)$$

$$M\dot{\omega} = \kappa P_{ref} - P_{inv} - D_v(\omega - 1), \quad (22)$$

$$\tau_E \dot{E} = k_Q(\kappa Q_{ref} - Q_{inv}) - k_E(E - E_{ref}). \quad (23)$$

Equation (23) is the decisive correction. Replacing E_{ref} by measured $|V|$ would remove the intended external voltage-restoration term at equilibrium and changes the small-signal physics. The voltage reference is

$$v_d^* = E, \quad v_q^* = 0, \quad qquad i_d^* = k_{pV}(v_d^* - v_d) + k_{iV}\xi_{Vd}, \quad i_q^* = k_{pV}(v_q^* - v_q) + k_{iV}\xi_{Vq}. \quad (24)$$

The limited current reference then enters the same physically signed inner current controller and command-delay plant. Here the limiter-owning states are the voltage-loop integrators:

$$\dot{\xi}_{Vd} = \begin{cases} 0, & \alpha < 1 \text{ and } k_{iV}e_{Vd}r_d > 0, \\ e_{Vd}, & \text{otherwise,} \end{cases} \quad \dot{\xi}_{Vq} = \begin{cases} 0, & \alpha < 1 \text{ and } k_{iV}e_{Vq}r_q > 0, \\ e_{Vq}, & \text{otherwise.} \end{cases} \quad (25)$$

The inner states still satisfy $\dot{\xi}_{Id} = e_d$, $\dot{\xi}_{Iq} = e_q$. No PLL state or hidden PLL angle is used in this branch.

6 Equilibrium, references, and SSSA

At SG trip the full nonlinear equilibrium solves every rectangular KCL row,

$$YV - I_{inj}(x, V, u) = 0, \quad (26)$$

plus one $\dot{E} = 0$ equation per selected GFM. Terminal GFM reactive power is a solved output; Q_{ref} and E_{ref} remain immutable case/event inputs. One scalar active-power mismatch is distributed by the frozen participation contract. A newly forming GFM holds its pre-trip P_{ref} when GFL resources remain; those GFL resources share the lost-SG deficit by P_{max} . If every IBR is GFM, the generic post-trip participation is used because no GFL remains.

One angle coordinate per energized island is removed as a mathematical gauge. This does not mean that only one physical GFM may operate. Several GFMs may form the same island, but they share one common rotational invariance. SSSA retains all relative electromechanical and controller modes and quotients only that common VSG-angle direction; no eigenvalue is deleted after calculation.

7 How the number of GFMs is selected

For SG-off operation the automatic selector enumerates $n_{GFM} = 1, 2, 3, 4$, every subset of that size, and every admissible reference member. Each candidate must pass topology, full-KCL equilibrium, operating limits, and the physical SSSA margin $\Omega \leq -\gamma_{req}$. The frozen ranking is: feasible first, fewer event-left mode changes, fewer GFMs, larger damping margin, deterministic resource-ID tie-break. Thus the current policy means *minimum small-signal-feasible formation*, not maximum transient robustness.

For the corrected case, 15 candidates were evaluated. Composite resource indices are 1 =SG1, 2 =IBR2, 3 =IBR3, 4 =IBR6, and 5 =IBR8. Two candidates passed:

Selected resources	n_{GFM}	Reference	Ω (s ⁻¹)	KCL norm
[5]	1	5	-0.175750	5.311×10^{-13}
[2, 3, 4, 5]	4	2	-2.344851	4.025×10^{-12}

The four-GFM point has the stronger local damping margin and more distributed voltage-forming support. The one-GFM point changes fewer devices and permits a bumpless active-power handover in which the remaining GFLs absorb the lost-SG deficit. Neither SSSA result alone proves survival of the complete large disturbance; the 200-s chronology is the falsification test.

7.1 Complete selector and SSSA mode inventory

The following table reports all 15 SG-off candidates. “Eq.” means that the full nonlinear all-KCL equilibrium stage was attempted; “SSSA” means that the candidate reached the full-state linearisation. A failed candidate is retained as falsification evidence and is not promoted to a readiness result.

n_{GFM}	Selected	Ref.	Eq.	SSSA	Ω (s ⁻¹)	Margin	Result
1	5	5	yes	yes	-0.175750	0.075750	PASS
1	3	3	yes	yes	1.059319	-1.159319	SSSA: insufficient margin
1	4	4	yes	yes	2.050660	-2.150660	SSSA: insufficient margin
1	2	2	yes	yes	3.692245	-3.792245	SSSA: insufficient margin
2	[3 4]	3	yes	yes	4.442894	-4.542894	SSSA: insufficient margin
2	[2 4]	2	yes	yes	5.275795	-5.375795	SSSA: insufficient margin
2	[2 3]	2	yes	no	–	–	equilibrium: device limit
2	[2 5]	2	yes	no	–	–	equilibrium: device limit
2	[3 5]	3	yes	no	–	–	equilibrium: device limit
2	[4 5]	4	yes	no	–	–	equilibrium: device limit
3	[2 3 4]	2	yes	no	–	–	equilibrium: no convergence
3	[2 3 5]	2	yes	no	–	–	equilibrium: no convergence
3	[2 4 5]	2	yes	no	–	–	equilibrium: no convergence
3	[3 4 5]	3	yes	no	–	–	equilibrium: no convergence
4	[2 3 4 5]	2	yes	yes	-2.344851	2.244851	PASS

For a physical decision matrix A_p , right eigenvectors R and the paired left-eigenvector rows L satisfy

$$A_p R = R \Lambda, \quad L = R \setminus I, \quad \pi_{ki} = \frac{|R_{ki} L_{ik}|}{\sum_j |R_{ji} L_{ij}|}. \quad (27)$$

The participation values below are reporting diagnostics; they never feed candidate ranking. Each complex-conjugate pair is printed once, with modal frequency $f_i = |\Im \lambda_i|/(2\pi)$ and damping ratio $\zeta_i = -\Re \lambda_i/|\lambda_i|$. The owner is shown explicitly as device:mathematical-state-symbol. For repeated roots, especially the four exact -10 s^{-1} dc-link roots, individual eigenvectors may rotate inside the repeated eigenspace; they should be interpreted as one dc-link family rather than four uniquely attributable modes.

Automatic winner: one GFM

The active inventory is one 12-state GFM at resource 5 plus three 11-state GFLs, giving 45 full active states. Quotienting the common VSG angle gives 44 physical decision modes. The table has 28 rows because conjugate pairs are shown once. The limiting pair is $-0.175750 \pm j2.159250 \text{ s}^{-1}$ (0.343655 Hz), dominated by the three remaining GFL PLL-angle states.

No.	Eigenvalue $\lambda \text{ (s}^{-1}\text{)}$	$f \text{ (Hz)}$	ζ	Dominant participation: device:state (normalised)
1	$-3881.470073 + j1061.916124$	+ 169.009200	0.964553	IBR6: i_d (0.131); IBR6: i_q (0.131); IBR8: i_d (0.129)
2	-159.046981	0	1.000000	IBR8: E^{GFM} (0.992); IBR8: $V_{d,del}^{GFM}$ (0.006); IBR8: ξ_{Id}^{GFM} (0.001)
3	$-156.112829 + j816.094065$	+ 129.885404	0.187886	IBR6: i_d (0.305); IBR6: i_q (0.305); IBR3: i_d (0.105)
4	$-140.724047 + j1033.942353$	+ 164.557036	0.134861	IBR8: i_q (0.344); IBR8: i_d (0.343); IBR3: i_d (0.062)
5	$-106.207976 + j615.792052$	+ 98.006349	0.169964	IBR2: i_d (0.248); IBR2: i_q (0.248); IBR3: i_d (0.168)
6	-17.922231	0	1.000000	IBR8: ω_{VSG}^{GFM} (0.909); IBR8: ξ_{Id}^{GFM} (0.017); IBR6: ξ_{Id}^{GFL} (0.011)
7	$-13.345138 + j1.527324$	0.243081	0.993514	IBR2: ξ_{Id}^{GFL} (0.261); IBR2: ξ_{Iq}^{GFL} (0.261); IBR3: ξ_{Iq}^{GFL} (0.171)
8	$-13.151005 + j2.154763$	0.342941	0.986841	IBR6: ξ_{Id}^{GFL} (0.281); IBR6: ξ_{Iq}^{GFL} (0.280); IBR3: ξ_{Iq}^{GFL} (0.106)
9	$-12.633262 + j3.598177$	0.572668	0.961751	IBR8: ξ_{Id}^{GFM} (0.227); IBR8: ξ_{Iq}^{GFM} (0.223); IBR8: $V_{d,del}^{GFM}$ (0.062)
10	$-11.446947 + j14.627753$	2.328079	0.616280	IBR8: $V_{d,del}^{GFM}$ (0.155); IBR8: ξ_{Id}^{GFM} (0.142); IBR6: $V_{d,del}^{GFL}$ (0.099)
11	-10.000000	0	1.000000	IBR2: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
12	-10.000000	0	1.000000	IBR3: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
13	-10.000000	0	1.000000	IBR6: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
14	-10.000000	0	1.000000	IBR8: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
15	$-6.988836 + j12.831357$	2.042174	0.478320	IBR8: $V_{q,del}^{GFM}$ (0.155); IBR8: ξ_{Iq}^{GFM} (0.145); IBR6: $V_{q,del}^{GFL}$ (0.094)

No.	Eigenvalue λ (s ⁻¹)	f (Hz)	ζ	Dominant participation: device:state (normalised)
16	-6.293824 + j109.583760	17.440797	0.057339	IBR2: $V_{d,del}^{GFL}$ (0.224); IBR2: $V_{q,del}^{GFL}$ (0.223); IBR3: $V_{q,del}^{GFL}$ (0.146)
17	-2.508684 + j84.237836	13.406868	0.029768	IBR6: $V_{d,del}^{GFL}$ (0.256); IBR6: $V_{q,del}^{GFL}$ (0.253); IBR3: $V_{q,del}^{GFL}$ (0.101)
18	-2.048337	0	1.000000	IBR6: ξ_P^{GFL} (0.289); IBR2: ξ_P^{GFL} (0.235); IBR3: ξ_P^{GFL} (0.212)
19	-1.643720 + j47.532682	7.565061	0.034560	IBR8: $V_{d,del}^{GFM}$ (0.219); IBR8: $V_{q,del}^{GFM}$ (0.214); IBR8: ξ_{Id}^{GFM} (0.060)
20	-1.509781	0	1.000000	IBR6: ξ_Q^{GFL} (0.321); IBR6: ξ_P^{GFL} (0.265); IBR3: ξ_P^{GFL} (0.128)
21	-1.503650 + j0.674893	0.107413	0.912318	IBR8: ξ_{Vq}^{GFM} (0.286); IBR6: ξ_Q^{GFL} (0.122); IBR8: ξ_{Vd}^{GFM} (0.111)
22	-1.443329	0	1.000000	IBR3: ξ_P^{GFL} (0.299); IBR2: ξ_P^{GFL} (0.275); IBR2: ξ_Q^{GFL} (0.227)
23	-1.407957	0	1.000000	IBR2: ξ_Q^{GFL} (0.343); IBR2: ξ_P^{GFL} (0.288); IBR6: ξ_P^{GFL} (0.157)
24	-1.390336	0	1.000000	IBR3: ξ_Q^{GFL} (0.376); IBR3: ξ_P^{GFL} (0.246); IBR6: ξ_P^{GFL} (0.176)
25	-0.746742	0	1.000000	IBR8: ξ_{Vd}^{GFM} (0.624); IBR6: ξ_P^{GFL} (0.087); IBR2: ξ_P^{GFL} (0.082)
26	-0.619437 + j2.216829	0.352819	0.269116	IBR6: δ_{PLL}^{GFL} (0.278); IBR6: ξ_{PLL}^{GFL} (0.277); IBR2: δ_{PLL}^{GFL} (0.196)
27	-0.605533 + j2.186111	0.347930	0.266940	IBR3: δ_{PLL}^{GFL} (0.305); IBR3: ξ_{PLL}^{GFL} (0.304); IBR2: δ_{PLL}^{GFL} (0.149)
28	-0.175750 + j2.159250	0.343655	0.081126	IBR3: δ_{PLL}^{GFL} (0.136); IBR2: δ_{PLL}^{GFL} (0.133); IBR6: δ_{PLL}^{GFL} (0.128)

Alternative feasible point: four GFMs

Four 12-state GFMs give 48 full active states and 47 physical decision modes after the same one-angle quotient. The table has 30 rows with conjugate pairs printed once. No PLL state exists in this configuration. The limiting root is the real mode -2.344851 s⁻¹, dominated by GFM voltage-loop integrators, while the principal VSG angle/frequency families appear at approximately 17.97–23.95 Hz.

No.	Eigenvalue λ (s ⁻¹)	f (Hz)	ζ	Dominant participation: device:state (normalised)
1	-3874.026967 + j1062.858968	+ 169.159259	0.964364	IBR6: i_q (0.131); IBR6: i_d (0.131); IBR8: i_q (0.128)
2	-160.751647 + j771.363375	+ 122.766294	0.204016	IBR6: i_q (0.306); IBR6: i_d (0.305); IBR3: i_q (0.108)
3	-159.870515	0	1.000000	IBR2: E^{GFM} (0.257); IBR8: E^{GFM} (0.254); IBR6: E^{GFM} (0.245)

No.	Eigenvalue λ (s ⁻¹)	f (Hz)	ζ	Dominant participation: device:state (normalised)
4	-158.678405	0	1.000000	IBR8: E^{GFM} (0.709); IBR3: E^{GFM} (0.156); IBR2: E^{GFM} (0.108)
5	-158.281882	0	1.000000	IBR6: E^{GFM} (0.710); IBR3: E^{GFM} (0.199); IBR2: E^{GFM} (0.049)
6	-157.863132	0	1.000000	IBR2: E^{GFM} (0.573); IBR3: E^{GFM} (0.390); IBR6: E^{GFM} (0.018)
7	-142.863696 + j1020.567960	+ 162.428436	0.138633	IBR8: i_q (0.343); IBR8: i_d (0.342); IBR3: i_q (0.061)
8	-118.069613 + j555.888842	+ 88.472457	0.207763	IBR2: i_q (0.246); IBR2: i_d (0.245); IBR3: i_q (0.165)
9	-18.034003	0	1.000000	IBR8: ω_{VSG}^{GFM} (0.257); IBR6: ω_{VSG}^{GFM} (0.224); IBR2: ω_{VSG}^{GFM} (0.213)
10	-12.517483 + j16.868689	2.684735	0.595908	IBR2: $V_{d,del}^{GFM}$ (0.079); IBR3: $V_{q,del}^{GFM}$ (0.074); IBR2: $V_{q,del}^{GFM}$ (0.074)
11	-11.633833 + j13.272985	2.112461	0.659145	IBR3: ξ_{Iq}^{GFM} (0.087); IBR3: $V_{q,del}^{GFM}$ (0.087); IBR2: ξ_{Id}^{GFM} (0.084)
12	-10.000000	0	1.000000	IBR2: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
13	-10.000000	0	1.000000	IBR3: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
14	-10.000000	0	1.000000	IBR6: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
15	-10.000000	0	1.000000	IBR8: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
16	-8.750654 + j14.210463	2.261665	0.524347	IBR3: $V_{q,del}^{GFM}$ (0.176); IBR2: $V_{q,del}^{GFM}$ (0.170); IBR3: ξ_{Iq}^{GFM} (0.158)
17	-7.899977 + j14.415799	2.294346	0.480577	IBR8: $V_{q,del}^{GFM}$ (0.192); IBR8: ξ_{Iq}^{GFM} (0.166); IBR2: $V_{q,del}^{GFM}$ (0.051)
18	-7.522961 + j130.919399	20.836470	0.057368	IBR6: δ_{VSG}^{GFM} (0.353); IBR6: ω_{VSG}^{GFM} (0.299); IBR3: ω_{VSG}^{GFM} (0.096)
19	-7.491432 + j112.918894	17.971600	0.066198	IBR8: δ_{VSG}^{GFM} (0.409); IBR8: ω_{VSG}^{GFM} (0.299); IBR3: ω_{VSG}^{GFM} (0.078)
20	-6.973071 + j9.271486	1.475603	0.601073	IBR6: ξ_{Iq}^{GFM} (0.147); IBR6: ξ_{Id}^{GFM} (0.144); IBR6: $V_{q,del}^{GFM}$ (0.125)
21	-5.618627 + j17.153529	2.730069	0.311276	IBR6: $V_{q,del}^{GFM}$ (0.194); IBR6: ξ_{Iq}^{GFM} (0.137); IBR6: $V_{d,del}^{GFM}$ (0.104)
22	-4.987193 + j150.493385	23.951766	0.033121	IBR3: δ_{VSG}^{GFM} (0.321); IBR2: ω_{VSG}^{GFM} (0.261); IBR3: ω_{VSG}^{GFM} (0.134)
23	-4.598028 + j11.579746	1.842974	0.369046	IBR8: $V_{d,del}^{GFM}$ (0.127); IBR8: ξ_{Id}^{GFM} (0.117); IBR8: $V_{q,del}^{GFM}$ (0.065)
24	-3.635015	0	1.000000	IBR2: ξ_{Vd}^{GFM} (0.362); IBR3: ξ_{Vd}^{GFM} (0.200); IBR3: ξ_{Vq}^{GFM} (0.117)
25	-3.360497 + j0.551305	0.087743	0.986809	IBR8: ξ_{Vq}^{GFM} (0.313); IBR8: ξ_{Vd}^{GFM} (0.244); IBR2: ξ_{Vd}^{GFM} (0.087)

No.	Eigenvalue λ (s^{-1})	f (Hz)	ζ	Dominant participation: device:state (normalised)
26	-2.880396	0	1.000000	IBR2: ξ_{Vd}^{GFM} (0.325); IBR3: ξ_{Vq}^{GFM} (0.264); IBR2: ξ_{Vq}^{GFM} (0.223)
27	-2.828050 + j1.409490	0.224327	0.895001	IBR6: ξ_{Vq}^{GFM} (0.344); IBR6: ξ_{Vd}^{GFM} (0.300); IBR8: ξ_{Vd}^{GFM} (0.097)
28	-2.503201 + j9.251750	1.472462	0.261174	IBR2: $V_{d,del}^{GFM}$ (0.139); IBR2: ξ_{Id}^{GFM} (0.129); IBR3: δ_{VSG}^{GFM} (0.114)
29	-2.431408	0	1.000000	IBR2: ξ_{Vd}^{GFM} (0.395); IBR2: ξ_{Vq}^{GFM} (0.239); IBR3: ξ_{Vq}^{GFM} (0.179)
30	-2.344851	0	1.000000	IBR2: ξ_{Vq}^{GFM} (0.410); IBR2: ξ_{Vd}^{GFM} (0.184); IBR3: ξ_{Vd}^{GFM} (0.111)

The exact physical coordinate inventories, including device and controller owner for every coordinate, are retained as generated audit tables:

- one-GFM inventory: figures/switch_ieee14/sssa_states_n1.tex;
- four-GFM inventory: figures/switch_ieee14/sssa_states_n4.tex.

8 Hybrid event and integration contract

At a mode transaction, the device reset map preserves the terminal port (V, I, P, Q) and V_{dc} , initializes only the target controller states, then solves the exact right-limit full-KCL algebraic system. A failure rolls the whole transaction back. Reference ownership is supervisory and physical; it never replaces or removes a KCL row.

Normal continuous steps use coupled implicit trapezoidal integration:

$$x_{k+1} - x_k - \frac{h}{2} [f(x_k, y_k) + f(x_{k+1}, y_{k+1})] = 0, qquad g(x_{k+1}, y_{k+1}) = 0. \quad (28)$$

After a discontinuity, only the first interval is restarted with two backward-Euler half steps,

$$x_{m+1} - x_m - \frac{h}{2} f(x_{m+1}, y_{m+1}) = 0, \quad (29)$$

before returning to trapezoidal integration. This Rannacher restart damps the nonphysical step-to-step ringing of a stiff trapezoidal discretisation; it does not smooth, clip, or modify the physical trajectory.

If a logical output step requires dyadic subdivision, the next logical step starts from the most recently accepted subdivision level and then coarsens when the full step becomes feasible again. Every leaf still solves the identical implicit residual with the identical Newton tolerance. This history is only an initial step-partition choice: it changes neither the differential/algebraic equations nor the acceptance gate, and avoids repeatedly retrying coarse ancestors already known to fail in the fault and handback transients.

The per-device severity used for release contains exactly the requested voltage and frequency terms:

$$J_{V,i} = \frac{||V_i| - |V_{i,healthy}||}{0.10}, \quad J_f = \frac{|f_{COI} - 60|}{0.50}, \quad S_i = \text{sat}_{[0,1]}(0.5J_{V,i} + 0.5J_f). \quad (30)$$

The healthy voltage is the case-defined SG-online PF value at that IBR bus. RoCoF, active-power error, SCR, PLL-lock, and reference-availability diagnostics do not enter this scalar severity; applicable hard admissibility checks remain separate.

While the SG is offline the same two-term signal is evaluated in real time. If the system severity remains at or above $\Gamma_{on} = 0.65$ for $T_{d,on} = 0.10$ s, the selector commits the smallest authenticated feasible strict superset of the present GFM set. If it remains below $\Gamma_{off} = 0.35$ for $T_{d,off} = 1.00$ s, it may commit a feasible subset, but at least one GFM and one authenticated island reference are retained. In the generated run, SG trip initially committed one GFM and the measured stress added three resources at 21.2 s; this is an event-driven one-to-four transition, not a manual four-GFM schedule.

The breaker-open SG is synchronized to the live GFM-formed grid rather than to nominal zero slip. With $T_e = 0$, define

$$2H\dot{\omega}_{SG} = P_m - D\omega_{SG}, \quad \dot{\delta}_{SG} = \omega_0\omega_{SG}, \quad (31)$$

$$e_\theta = \text{wrap}(\angle V_{bus} - \angle V_{SG,oc}), \quad e_\omega = \omega_g - \omega_{SG}. \quad (32)$$

The project-derived mechanical command is

$$P_c = \text{sat}_{[0, P_{max}]} (D\omega_g + K_\omega e_\omega + K_\theta e_\theta), \quad (33)$$

$$K_\omega = 4H\zeta\omega_n - D, \quad qquad K_\theta = \frac{2H\omega_n^2}{\omega_0}, \quad (34)$$

$$T_{sv}\dot{P}_{sv} = -P_{sv} + P_c, \quad qquad T_{ch}\dot{P}_m = -P_m + P_{sv}, \quad (35)$$

with $\omega_n = 0.8$ rad/s, $\zeta = 1$, $T_{sv} = 0.2$ s, and $T_{ch} = 0.4$ s. The voltage matcher acts separately on field input. The close request at 145 s is only the earliest allowed time; the case provides a 20-s synchronization opportunity and retains the unchanged 0.5-s dwell. The breaker closed at 154.300 s with $\Delta V = 0.018675$ pu, $\Delta f = 0.000107$ pu, and $\Delta\theta = 5.612123^\circ$.

The post-reclose handback is not an all-device hard-coded command. SG must first pass the synchronism dwell. Thereafter each remaining GFM independently requires $S_i < \Gamma_{off}$ for $T_{d,off}$, followed by its own transfer and KCL transaction. The generated run records releases at 159.200, 162.700, 193.000, and 194.100 s; all four IBRs are GFL from 194.100 s onward.

9 Generated event and trajectory evidence

Time (s)	Event / index	Timer	Mode before after	→	Reason
20.000	SG trip / 0.000	reference-loss guard	all GFL GFM / 3 GFL	→ 1	SG opened; authenticated selected-GFM transaction committed
21.200	GFM support add / 0.708	0.1-s severity dwell	1 GFM / 3 GFL → all GFM		SG-off severity high; feasible strict superset committed
50.000	load step / 1.000	–	all GFM GFM	→ all	base constant-impedance load increased
85.000	fault on / 1.000	–	all GFM GFM	→ all	bus-9 shunt fault applied
85.150	fault clear / 1.000	–	all GFM GFM	→ all	fault admittance removed
110.000	line trip / 1.000	–	all GFM GFM	→ all	line 6–13 opened
145.000	topology restore / 1.000	–	all GFM GFM	→ all	base load and line restored
145.000	SG close request / 1.000	–	all GFM GFM	→ all	earliest close request; guard monitored
154.300	SG reclose / 1.000	0.5-s sync dwell	all GFM GFM	→ all	guard passed; SG owns reference; IBR modes unchanged
159.200	severity release / 0.299	1.0-s severity dwell	all GFM GFM / 1 GFL	→ 3	two-term severity dwell passed; transfer/KCL committed
162.700	severity release / 0.583	1.0-s severity dwell	3 GFM / 1 GFL → 2 GFM / 2 GFL		two-term severity dwell passed; transfer/KCL committed
193.000	severity release / 0.160	1.0-s severity dwell	2 GFM / 2 GFL → 1 GFM / 3 GFL		two-term severity dwell passed; transfer/KCL committed
194.100	severity release / 0.291	1.0-s severity dwell	1 GFM / 3 GFL → all GFL		two-term severity dwell passed; transfer/KCL committed

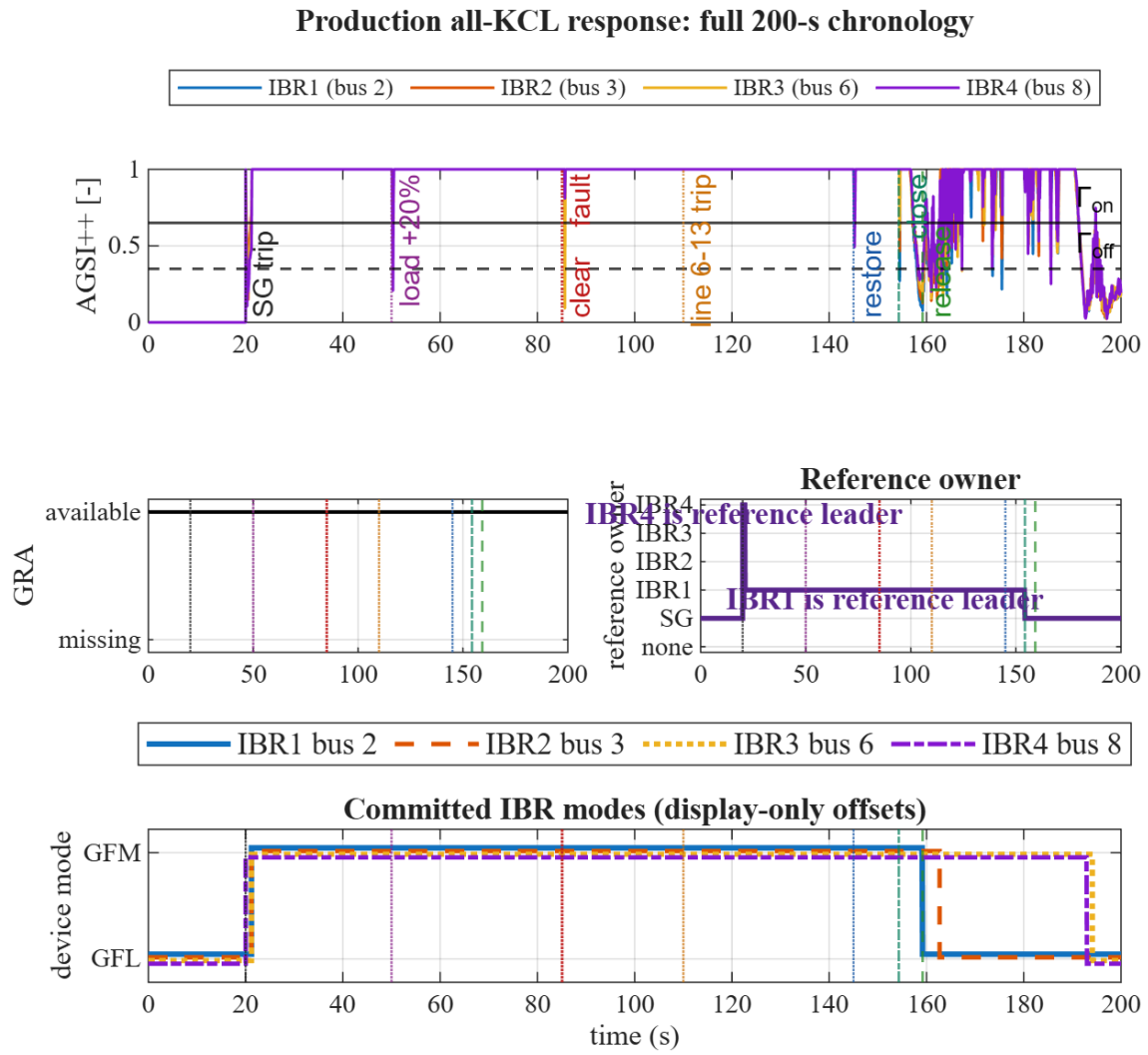


Figure 1: Raw accepted severity, reference ownership, and committed GFL/GFM modes.

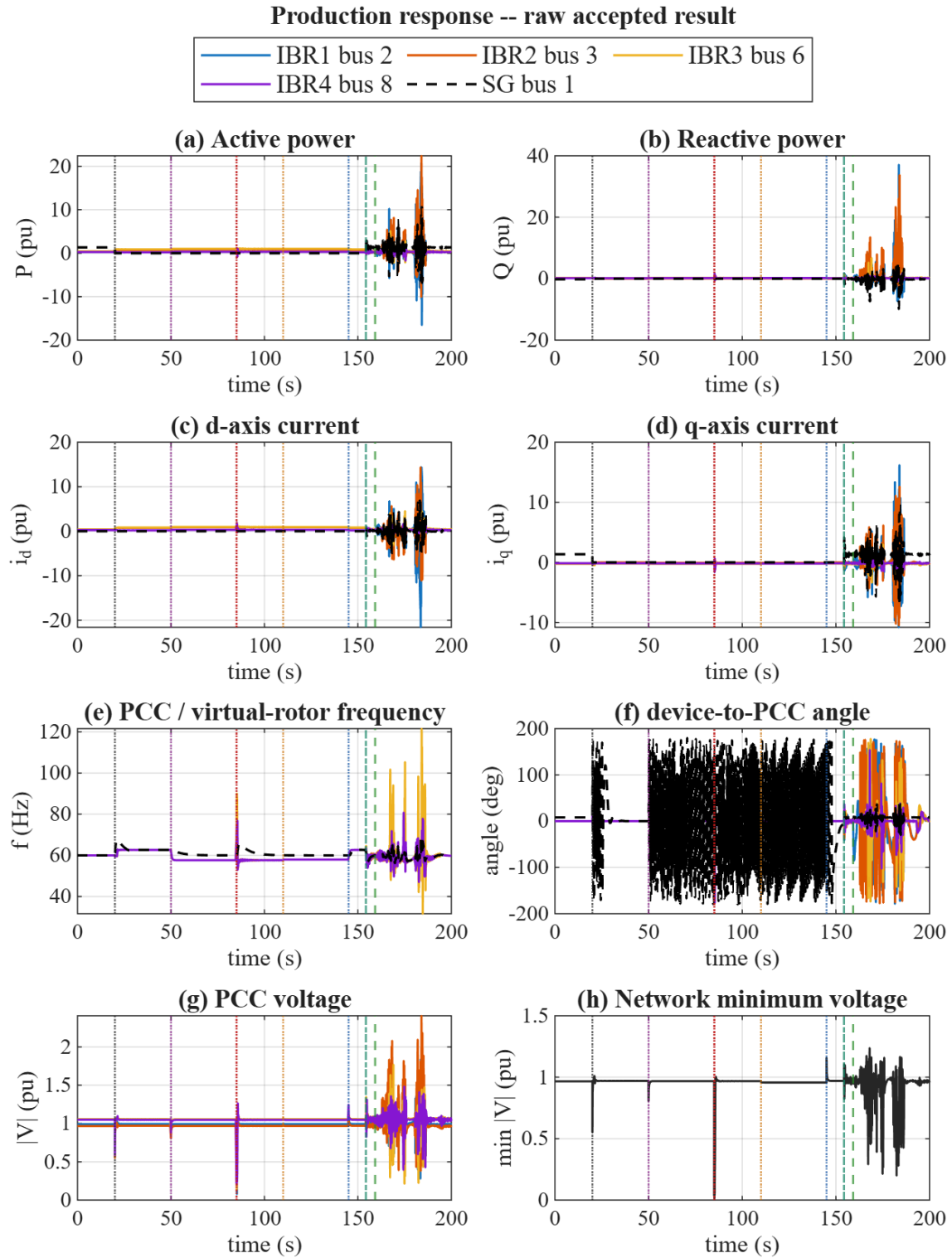


Figure 2: Raw electrical trajectory from the accepted production result; no smoothing, clipping, filtering, or synthetic overlay is applied.

The observed voltage and frequency extrema are deliberately not acceptance thresholds. A finite trajectory and converged all-KCL steps establish numerical completion of the requested run, while the excursion remains a model/controller limitation for subsequent development.

10 Verification and reproduction

Independent targeted verification covers: fixed state ownership; GFL PLL sensitivity; GFM no-PLL angle ownership; the exact E_{ref} coefficient in (23); dc restoration; terminal-port continuity; per-unit power covariance; source-model SCR applicability; full-KCL equilibrium; physical SSSA; default trapezoidal equivalence; backward-Euler closed-form behavior; event rollback; SG reclose; and index-driven handback. The focused regression result is 71/71 passed. The full repository suite was intentionally omitted under the repository risk policy; the targeted set covers the changed producer, consumers, and event failure paths.

Reproduction commands are:

```
matlab -batch "addpath('scripts/reporting'); ...
    generate_gfl_gfm_sssa_tables"
matlab -batch "addpath('scripts/diagnostics'); ...
    run_ieee14_gfl_gfm_state_release"
matlab -batch "addpath('scripts/reporting'); ...
    generate_ieee14_switch_report_figures(reuse_cache=true)"
```

The first command rebuilds the authenticated selector and all SSSA tables. The second command writes continuous progress and the complete transient result to:

- output/diagnostics/engine_release_progress.log
- output/diagnostics/engine_release_result.log
- output/diagnostics/engine_release_result.mat

The third command regenerates the figures, event table, and numerical macros used by this report.

11 Limitations

- The dc-source controller is a declared project-derived closure because the case does not prescribe the external dc current law.
- The automatic selector is exhaustive only for at most four eligible IBRs. Larger systems fail closed pending a validated scalable search.
- Candidate feasibility includes full P/Q balance and local operating limits at equilibrium, but it does not yet include a time-domain reserve/security score in its ranking.
- The model is balanced positive-sequence. Unbalanced and switching-frequency behavior is outside scope.
- Transient voltage oscillation is reported honestly and is not a readiness claim.